

Defects

```
library(mosaic)

## Registered S3 method overwritten by 'mosaic':
##   method                                from
##   fortify.SpatialPolygonsDataFrame ggplot2

##
## The 'mosaic' package masks several functions from core packages in
## order to add
## additional features. The original behavior of these functions
## should not be affected by this.

##
## Attaching package: 'mosaic'

## The following objects are masked from 'package:dplyr':
##
##   count, do, tally

## The following object is masked from 'package:Matrix':
##
##   mean

## The following object is masked from 'package:ggplot2':
##
##   stat

## The following objects are masked from 'package:stats':
##
##   binom.test, cor, cor.test, cov, fivenum, IQR, median,
##   prop.test,
##   quantile, sd, t.test, var

## The following objects are masked from 'package:base':
##
##   max, mean, min, prod, range, sample, sum

library(dplyr)
library(ggplot2)
library(lubridate)
```

```
##
## Attaching package: 'lubridate'

## The following objects are masked from 'package:base':
##
##     date, intersect, setdiff, union

library(Matrix)
library(olsrr)

##
## Attaching package: 'olsrr'

## The following object is masked from 'package:datasets':
##
##     rivers

library(readr)
library(lattice)
library(readxl)
library(scales)

##
## Attaching package: 'scales'

## The following object is masked from 'package:readr':
##
##     col_factor

## The following object is masked from 'package:mosaic':
##
##     rescale

library(janitor)

##
## Attaching package: 'janitor'

## The following objects are masked from 'package:stats':
##
##     chisq.test, fisher.test

Defects <- read_excel("~/Documents/Defects/Defects_Dataset.xlsx")
```

```

# defects counts
names(Defects)

## [1] "Reference_Number"      "Investigation_ID"
"Transmission_Downshift"
## [4] "Timing_Belt_Failure"   "B_Pillar_Trim"
"Defect_Count"

data.frame(
  Defect_Type = c("Transmission Downshift", "Timing Belt Failure", "B-
Pillar Trim"),
  Count = colSums(Defects[, c("Transmission_Downshift",
                              "Timing_Belt_Failure",
                              "B_Pillar_Trim")])
)

##                               Defect_Type Count
## Transmission_Downshift Transmission Downshift    329
## Timing_Belt_Failure      Timing Belt Failure    44
## B_Pillar_Trim              B-Pillar Trim        5

# calculating total number of defects
defects_total <- sum(Defects$Defect_Count)

# number of defects in dataset
units <- nrow(Defects)

# number of possible defects
opportunities <- 3

# defects per million opportunities (DPMO), standardizing defects rate
to "per million" scale
dpmo <- (defects_total / (units * opportunities)) * 1000000
dpmo

## [1] 333333.3

# converting DPMO to sigma level
sigma <- qnorm(1 - dpmo/1000000) + 1.5
sigma

## [1] 1.930727

```

Out of 378 defects * 3 opportunities per unit = 1134 chances for defects. Therefore the defect rate is 33.3%. If we look at it per million, we have 333,333 per 1,000,000 chances of defects, which indicates that there is a very substantial amount of defects. This yields to a estimated performance of ~2.0 sigma, indicating poor manufacturing quality.

```
# counting total number of defects for each type
counts <- colSums(Defects[, c("Transmission_Downshift",
"Timing_Belt_Failure", "B_Pillar_Trim")])

# total number of complains
units <- nrow(Defects)

# calculating DPMO
dpmo_by_type <- (counts / units) * 1000000

# converting DPMO value to sigma level
sigma_by_type <- qnorm(1 - dpmo_by_type/1000000) + 1.5

# combining results into a clean table
data.frame(
  Defect = names(counts),
  DPMO = dpmo_by_type,
  Sigma = sigma_by_type
)

##                                Defect      DPMO      Sigma
## Transmission_Downshift Transmission_Downshift 870370.37 0.3718564
## Timing_Belt_Failure      Timing_Belt_Failure 116402.12 2.6931663
## B_Pillar_Trim            B_Pillar_Trim    13227.51 3.7194662
```

Through defect concentration analysis, it shows that the majority of defects are from Transmission_Downshift. Therefore, if we were to focus on improving this category, it would also significantly improve overall sigma performance.

```
# defect counts
Transmission <- 329
Timing_Belt_Failure <- 44
B_Pillar_Trim <- 5

# total units and opportunities
units <- 378
opportunities_per_unit <- 3
```

```

total_opportunities <- units * opportunities_per_unit

# trying different improvement levels
reduction <- c(0, 0.25, 0.50, 0.75, 0.90)

# new transmission defects after improvement
new_transmission <- Transmission * (1 - reduction)

# new total defects
totalnum_defects <- new_transmission + Timing_Belt_Failure +
B_Pillar_Trim

# new defect rate
defect_rate <- totalnum_defects / total_opportunities

# DPMO
dpmo <- defect_rate * 1000000

# sigma level
sigma <- qnorm(1 - dpmo/1000000) + 1.5

# results into table
results <- data.frame(
  Transmission_Reduction = c("0%", "25%", "50%", "75%", "90%"),
  New_Transmission_Defects = round(new_transmission, 1),
  Total_Defects = round(totalnum_defects, 1),
  Defect_Rate = round(defect_rate, 3),
  DPMO = round(dpmo, 1),
  Sigma_Level = round(sigma, 2)
)

results

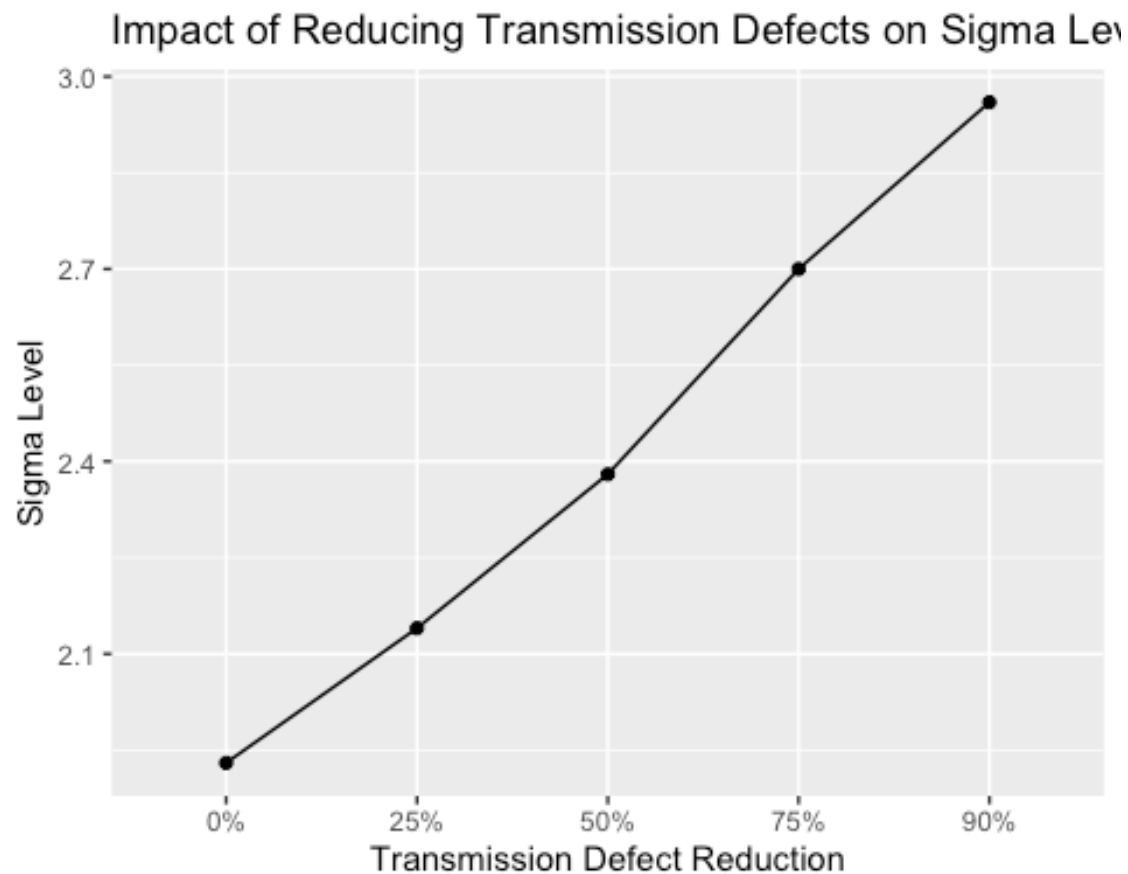
##   Transmission_Reduction New_Transmission_Defects Total_Defects
Defect_Rate
## 1                      0%                   329.0          378.0
0.333
## 2                      25%                   246.8          295.8
0.261
## 3                      50%                   164.5          213.5
0.188

```

## 4	75%	82.2	131.2
0.116			
## 5	90%	32.9	81.9
0.072			
##	DPMO	Sigma_Level	
## 1	333333.3	1.93	
## 2	260802.5	2.14	
## 3	188271.6	2.38	
## 4	115740.7	2.70	
## 5	72222.2	2.96	

As transmission defects are reduced, the total number of defects decreases significantly- leading to a lower defect rate and its corresponding improvement in the sigma level. This shows that transmission defects are the primary cause of poor manufacturing performance.

```
ggplot(results, aes(x = Transmission_Reduction, y = Sigma_Level, group
= 1)) +
  geom_line() +
  geom_point() +
  labs(
    title = "Impact of Reducing Transmission Defects on Sigma Level",
    x = "Transmission Defect Reduction",
    y = "Sigma Level"
  )
```



As transmission defects are reduced, the sigma level increases steadily, showing improved process throughout all three kinds of defects. Even moderate reductions in the main defect source lead to improvements in overall performance.

```
# x values
x <- seq(-6, 6, length = 1000)
y <- dnorm(x)

# sigma values
original_sigma <- 1.93
new_sigma <- 2.96

# empty plot
plot(
  x, y,
  type = "n",
```

```

main = "Sigma Distribution with Process Improvement",
xlab = "Sigma",
ylab = "Density",
ylim = c(0, 0.43)
)

# -----
# LEFT REGION: below 0 sigma
# -----
x_left <- x[x <= 0]
y_left <- dnorm(x_left)

polygon(
  c(x_left, rev(x_left)),
  c(y_left, rep(0, length(y_left))),
  col = "lightblue",
  border = NA
)

# -----
# MIDDLE REGION: 0 to 1.93 sigma
# -----
x_mid <- x[x >= 0 & x <= original_sigma]
y_mid <- dnorm(x_mid)

polygon(
  c(x_mid, rev(x_mid)),
  c(y_mid, rep(0, length(y_mid))),
  col = "lightgreen",
  border = NA
)

# -----
# IMPROVEMENT REGION:
# 1.93 sigma to 2.96 sigma
# -----
x_improve <- x[x >= original_sigma & x <= new_sigma]
y_improve <- dnorm(x_improve)

polygon(

```

```

c(x_improve, rev(x_improve)),
c(y_improve, rep(0, length(y_improve))),
col = "lightpink",
border = NA
)

# -----
# REMAINING TAIL:
# above 2.96 sigma
# -----
x_remaining <- x[x >= new_sigma]
y_remaining <- dnorm(x_remaining)

polygon(
  c(x_remaining, rev(x_remaining)),
  c(y_remaining, rep(0, length(y_remaining))),
  col = "darkseagreen3",
  border = NA
)

# redraw normal curve
lines(x, y, lwd = 2)

# sigma grid lines
abline(v = -6:6, lty = 3)

# original sigma line
abline(
  v = original_sigma,
  lty = 2,
  lwd = 2,
  col = "blue"
)

# new sigma line
abline(
  v = new_sigma,
  lty = 2,
  lwd = 2,
  col = "darkgreen"
)

```

```

)

# -----
# LABEL ORIGINAL SIGMA
# -----
text(
  original_sigma,
  0.40,
  labels = "1.93σ\nOriginal",
  pos = 2,
  col = "blue",
  font = 2,
  cex = 1.1
)

# -----
# LABEL NEW SIGMA
# -----
text(
  new_sigma,
  0.40,
  labels = "2.96σ\nAfter 90% Reduction",
  pos = 4,
  col = "darkgreen",
  font = 2,
  cex = 1.1
)

# -----
# PROCESS IMPROVEMENT ARROW
# -----
arrows(
  x0 = original_sigma,
  y0 = 0.33,
  x1 = new_sigma,
  y1 = 0.33,
  length = 0.12,
  lwd = 2
)

```

```
text(  
    (original_sigma + new_sigma) / 2,  
    0.35,  
    labels = "Process\nImprovement",  
    font = 2  
)
```

```
# -----  
# DEFECT REDUCTION LABEL  
# -----
```

```
arrows(  
    x0 = 2.55,  
    y0 = 0.14,  
    x1 = 2.35,  
    y1 = dnorm(2.35),  
    length = 0.1,  
    col = "red",  
    lwd = 2  
)
```

```
text(  
    2.65,  
    0.15,  
    labels = "Defects\nreduced",  
    col = "red",  
    font = 2  
)
```

```
# -----  
# REMAINING DEFECT LABEL  
# -----
```

```
arrows(  
    x0 = 3.75,  
    y0 = 0.07,  
    x1 = 3.35,  
    y1 = dnorm(3.35),  
    length = 0.1,  
    col = "darkgreen",  
    lwd = 2  
)
```

```

text(
  4.0,
  0.075,
  labels = "Remaining\ndefect rate\n(after 2.96σ)",
  col = "darkgreen",
  font = 2
)

# -----
# SIGMA LABELS -6 TO +6
# -----
for (i in -6:6) {
  text(
    i,
    0.015,
    paste0(i, "σ"),
    pos = 3
  )
}

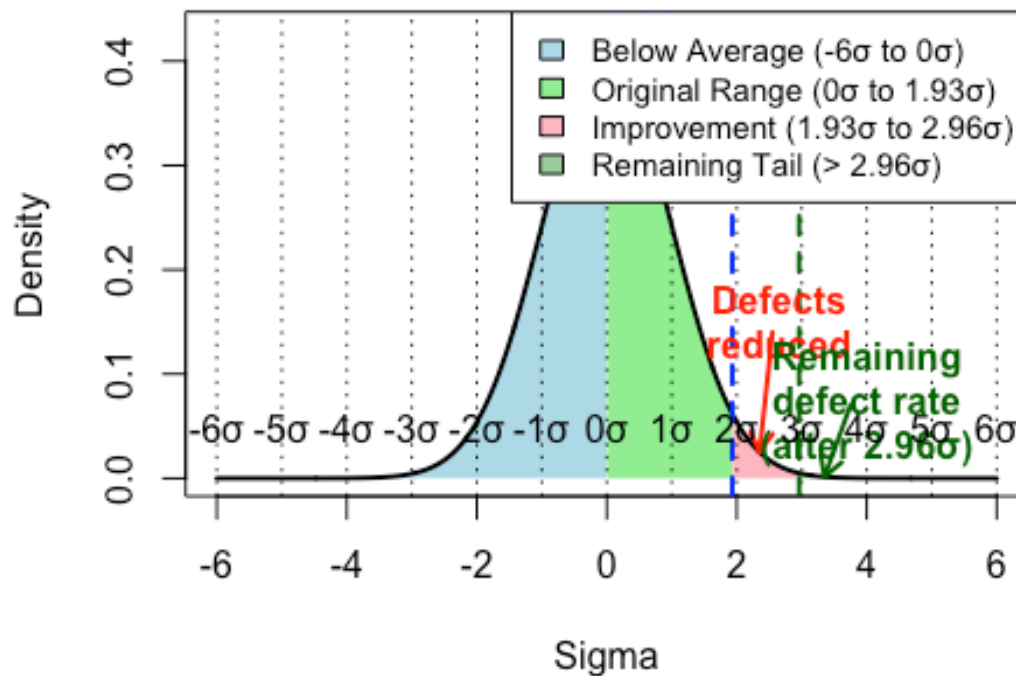
# -----
# LEGEND
# -----
legend(
  "topright",
  legend = c(
    "Below Average (-6σ to 0σ)",
    "Original Range (0σ to 1.93σ)",
    "Improvement (1.93σ to 2.96σ)",
    "Remaining Tail (> 2.96σ)"
  ),
  fill = c(
    "lightblue",
    "lightgreen",
    "lightpink",
    "darkseagreen3"
  ),
  border = "black",

```

cex = 0.85

)

Sigma Distribution with Process Improvement



The graph shows the Six Sigma range from -6 to $+6$. The original sigma level was 1.93σ , indicating relatively low process performance compared to the Six Sigma standard. After simulating a 90% reduction in transmission-downshift defects, the sigma level increased to 2.96σ , demonstrating a significant improvement in overall process quality. This increase shows that targeting the most common defect can reduce the overall defect rate and improve process performance. Further improvements at the manufacturing level, such as evaluating machinery, reducing human error, and implementing repeated quality testing, could potentially reduce defects even further and move the process closer to Six Sigma performance.